

## Earth's climate varies by latitude and season and is changing rapidly

The most significant influence on the distribution of organisms on land is **climate**, the long-term prevailing weather conditions in a given area. Four physical factors—temperature, precipitation, sunlight, and wind—are key components of climate. To set the stage for understanding how climate—and climate change—affect life on Earth, we'll begin by examining patterns in climate at the global, regional, and local levels.

### Global Climate Patterns

Global climate patterns are determined largely by the input of solar energy and Earth's movement in space. The sun warms the atmosphere, land, and water. This warming establishes the temperature variations, movements of air and water, and evaporation of water that cause dramatic latitudinal variations in climate. **Figure 52.3** summarizes Earth's climate patterns and how they are formed.

## Regional and Local Effects on Climate

Climate varies seasonally and can be modified by other factors, such as large bodies of water and mountain ranges. We will examine each of these factors in more detail.

### Seasonality

In middle to high latitudes, Earth's tilted axis of rotation and its annual passage around the sun cause strong seasonal cycles in day length, solar radiation, and temperature (**Figure 52.4**). The changing angle of the sun over the course of the year also affects local environments. For example, the belts of wet and dry air on either side of the equator move slightly northward and southward as the sun's angle changes; this produces marked wet and dry seasons around 20° north and 20° south latitude, where many tropical deciduous forests grow. In addition, seasonal changes in wind patterns alter ocean currents, sometimes causing the upwelling of cold water from deep ocean layers. This nutrient-rich water stimulates the growth of surface-dwelling phytoplankton and the organisms that feed on them. Though these upwelling zones make up only a few percent of the ocean's area, they are the source of more than 25% of all fish caught globally.

▼ **Figure 52.4 Seasonal variation in sunlight intensity.** Because Earth is tilted on its axis relative to its plane of orbit around the sun, the intensity of solar radiation varies seasonally. This variation is smallest in the tropics and increases toward the poles.

